

Week 01 Report

1. Introduction

The Housing dataset was analyzed to explore key factors affecting house prices, such as area, bedrooms, and overall price distribution. The main objective was to perform exploratory data analysis (EDA) to identify patterns, relationships, and insights using Python visualization tools.

2. Data Overview

The dataset was loaded and inspected using basic pandas functions such as `head()` and `info()`. It was found to contain both numerical and categorical features related to housing properties. A missing value check confirmed that the dataset had no null values, so no data cleaning for missing entries was required. Duplicate values were also checked to ensure data quality. Overall, the dataset was clean and ready for analysis.

3. Key Findings from EDA

3.1 Price Distribution: House prices showed a right-skewed distribution, meaning most houses fall in the lower to mid-price range, while only a few properties are highly expensive.

3.2 Relationship Between Area and Price: A clear positive relationship was observed between house area and price. Larger houses generally tend to have higher prices, although some variations and outliers exist.

3.3 Correlation Analysis: Correlation analysis indicated that area is one of the strongest factors influencing house price compared to other variables.

3.4 Outliers: The box plot revealed the presence of outliers in house prices, representing unusually expensive properties that may affect statistical analysis.

3.5 Bedroom Distribution: Most houses in the dataset have a moderate number of bedrooms, while very high-bedroom houses are rare, indicating typical residential patterns.